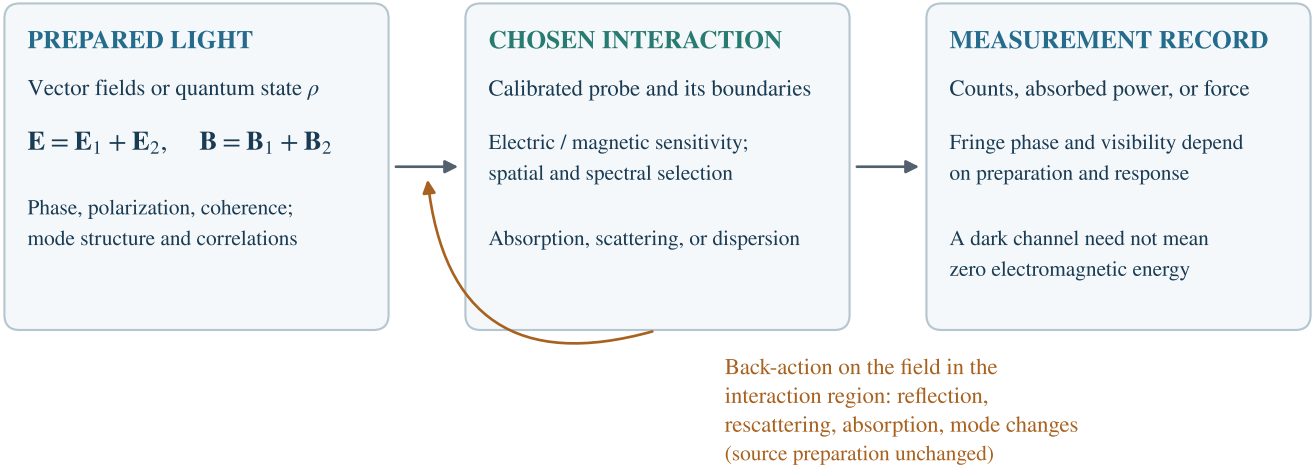
(a) Preparation, interaction, and recorded response



(b) Quantum description of a selected detection channel

$$\hat{b} = \kappa_1 \hat{a}_1 + \kappa_2 \hat{a}_2, \qquad R_b = \mathcal{K} \langle \hat{b}^\dagger \hat{b} \rangle$$

$$R_b/\mathcal{K} = |\kappa_1|^2 \langle \hat{n}_1 \rangle + |\kappa_2|^2 \langle \hat{n}_2 \rangle + 2\,\mathrm{Re}\left[\kappa_1^* \kappa_2 \langle \hat{a}_1^\dagger \hat{a}_2 \rangle\right]$$